

Inputs — calc

Design Calculation Sheet - V4b (VBF-T1R2-CALC-01) case t1_r2				
Parameter	Base (Chen2020)	V4b design value	Unit	Source
Positive electrode thickness	75.6	75.6	um	parameter set (base, unchanged)
Negative electrode thickness	85.2	96	um	cell/params_v4b.json
Separator thickness	12	9	um	cell/params_v4b.json
Positive current collector thickness	16	8	um	cell/params_v4b.json
Negative current collector thickness	12	6	um	cell/params_v4b.json
Positive electrode porosity	0.335	0.36	-	cell/params_v4b.json
Negative electrode porosity	0.25	0.32	-	cell/params_v4b.json
Separator porosity	0.47	0.5	-	cell/params_v4b.json
Positive particle radius	5.22	2	um	cell/params_v4b.json
Negative particle radius	5.86	1.5	um	cell/params_v4b.json
Positive electrode density	3262	3262	kg/m3	parameter set
Negative electrode density	1657	1657	kg/m3	parameter set
Separator density	397	397	kg/m3	parameter set
Positive CC density (Al)	2700	2700	kg/m3	parameter set
Negative CC density (Cu)	8960	8960	kg/m3	parameter set
Electrolyte conductivity	func(T)	1.8	S/m	bridge literature estimate (annotated)
Electrolyte diffusivity	func(T)	6e-10	m2/s	bridge literature estimate (annotated)
Cation transference number	0.2594	0.4	-	cell/params_v4b.json
Total heat transfer coefficient	10	60	W/m2.K	cell/params_v4b.json
Cell cooling surface area	0.00531	0.00531	m2	parameter set
Cell volume (thermal sim)	2.42e-05	2.514e-05	m3	cell/params_v4b.json (conservative scale, see design_spec.md § 6)
Electrode area	0.1027	0.1027	m2	cell/r4_v4b_energy.json:area_m2
Nominal capacity	5	5	Ah	parameter set
Upper voltage cut-off	4.2	4.2	V	parameter set
Lower voltage cut-off	2.5	2.5	V	parameter set

Capacity & Energy — calc

Quantity	Formula	Value	Source
1C discharge capacity	simulation (SPMe, 1C)	5.6997 Ah	cell/r4_v4b_1c_spme.json:capacity_ah
1C discharge energy	time integral $V \times I$	20.5534 Wh	cell/r4_v4b_energy.json:energy_wh
Midpoint voltage	mean of 1C voltage curve	4.0863 V	cell/r4_v4b_energy.json:midpoint_voltage_v
4C CC acceptance	$(t_{\text{end}} - t_{\text{vmin}}) \times 20 \text{ A} / 3600$	4.53 Ah (815 s)	inferred from cell/r4_v4b_4c_dfn.json voltage curve (capacity_ah key unreliable for charge protocols)
4C max temperature	DFN + lumped thermal, $h=60$	326.46 K	cell/r4_v4b_4c_dfn.json:T_max_K
4C anode potential min	DFN, vs Li/Li+	+0.0215 V -> plated false	cell/r4_v4b_4c_dfn.json:anode_potential_v (min)
Overcharge 4.7 V max temperature	0.5C charge to upper+0.5 V	299.43 K	cell/r4_v4b_overcharge.json:T_max_K
Thermal runaway trigger	run-tr, $mcp=31.72 \text{ J/K}$, $hA=0.3186$	false	cell/r4_v4b_tr.json:triggered

Energy Density — calc

Energy density chain (contract formula: ED = E_wh / mass_kg; mass = sum layer thickness x (1-porosity) x density x area; electrolyte excluded)				
Layer	layer_kg_m2 (kg/m2)	x area 0.1027 m2 -> kg	g/cell	Source
positive_electrode	0.157829	0.016209	16.209	cell/r4_v4b_energy.json:layer_kg_m2
negative_electrode	0.108169	0.011109	11.109	cell/r4_v4b_energy.json:layer_kg_m2
positive_cc	0.0216	0.002218	2.218	cell/r4_v4b_energy.json:layer_kg_m2
negative_cc	0.05376	0.005521	5.521	cell/r4_v4b_energy.json:layer_kg_m2
separator	0.001787	0.000184	0.183	cell/r4_v4b_energy.json:layer_kg_m2
TOTAL			35.241	
Quantity	Formula	Value	Unit	Source
Energy density	20.5534 Wh / 0.0352409 kg	583.227	Wh/kg	cell/r4_v4b_energy.json:energy_density_wh_kg
Volumetric energy density	E / volume	1028.422	Wh/L	cell/r4_v4b_energy.json:energy_density_wh_l
Energy density incl. electrolyte estimate	20.5534 Wh / 0.042935 kg	478.7	Wh/kg	inferred: +7.694 g electrolyte (1.2 g/cm3 literature estimate)
Criterion check	583.2 >= 392.61	PASS	-	entry-0 stage2 threshold; log-evaluate round 4

NP & Mass — calc

N/P ratio and mass breakdown			
Quantity	Formula	Value	Source
Positive areal mass	$75.6e-6 \times (1-0.36) \times 3262$	157.83 g/m ²	layer_kg_m2 x 1000
Negative areal mass	$96e-6 \times (1-0.32) \times 1657$	108.17 g/m ²	layer_kg_m2 x 1000
Positive areal capacity	157.83 g/m ² x 190 mAh/g	29.99 Ah/m ²	C_pos 190 mAh/g: NMC811 practical at 4.2 V, literature range 180-200, midpoint estimate
Negative areal capacity	108.17 g/m ² x 372 mAh/g	40.24 Ah/m ²	C_neg 372 mAh/g: graphite theoretical (literature)
N/P ratio	40.24 / 29.99	1.34 (quoted ~1.35; 1.31-1.38 across C_pos range)	consistent with propose R4 note 'N/P 1.35'
Cell mass (contract caliber)	sum layers	35.24 g	cell/r4_v4b_energy.json:mass_kg
Cell mass incl. electrolyte	+ 7.69 g	42.93 g	inferred

Process — calc

Process design parameters (consistent with design_spec.md § 3)				
Parameter	Formula	Value	Unit note	Source
Positive areal density	thickness x (1-porosity) x density	157.83	g/m2	layer_kg_m2 x 1000
Negative areal density	thickness x (1-porosity) x density	108.17	g/m2	layer_kg_m2 x 1000
Positive compaction density	3262 x (1-0.36) / 1000	2.09	g/cm3 (divide by 1000)	parameter-set density
Negative compaction density	1657 x (1-0.32) / 1000	1.13	g/cm3 (divide by 1000)	parameter-set density
Electrolyte fill amount	pore volume x 1.2 g/cm3 x fill 1.0	7.69	g/cell (pore volume 6.41 cm3)	literature density annotated estimate
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	-	design recommended value; production-line value requires tuning	-